

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with 6 years building event-driven, cloud-native services in Node.js and Python. Led the end-to-end, zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP — designing a real-time two-way sync pipeline on Google Pub/Sub that kept live traffic flowing throughout — and cut \$5,000/month in infrastructure costs. Hands-on with microservices, containerization (Docker, Kubernetes), MongoDB and SQL databases across AWS and GCP, with fintech exposure from real-time stock-market data systems and DeFi backends. Now applying the same full-SDLC ownership, from design through production operations, to backend services at a fully remote global bank.

CORE COMPETENCIES

Backend Services (Node.js, Python)

Event-Driven Architecture (Kafka, Pub/Sub, NATS)

Microservices & Containerization (Docker, Kubernetes)

MongoDB & SQL Databases

Cloud-Native: GCP & AWS

Zero-Downtime Migrations

Real-Time Financial Data Systems

Full SDLC Ownership

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer — US Remote

- Implemented **real-time two-way sync** between MongoDB and Firestore using Node.js and Google **Pub/Sub** for message processing, keeping both datastores consistent under live production traffic
- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, owning design through production operations and reporting directly to the CTO
- Saved the company **\$5,000/month** by migrating all services off AWS; moved file storage from Amazon S3 to GCS via a Python script
- Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion; implemented Express Shipping, generating an additional \$40 revenue per order

MTailor Mar 2022 – Jul 2022

Contractor (Backend Migration) — US Remote

- Built a migration pipeline reading from MongoDB and writing to Firestore after complex data-processing logic
- Wrote engineering documentation for the new Firebase SDKs (Node.js and Python) and trained the Ops team on dashboards and backup procedures

Dr Wealth Aug 2021 – Jan 2022

Software Engineer — Singapore Remote (Fintech)

- Led optimisation of a PWA delivering **real-time stock market prices** from a Firebase backend
- Built ad-hoc jobs querying over **2 million Firestore records** to keep customer prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js

Mind2matter.co Jan 2022 – Jul 2022

Software Engineer — US Remote

- Built backends for **DeFi applications** using Web3 and Node.js for a US agency with demanding clients
- Delivered React UIs from Figma designs under tight timelines

CodeBits Jan 2020 – Jul 2021

Software Engineer (Team Lead) — Uganda Remote

- Architected a **microservices** backend for FIDA Uganda's case management app using Apache **Kafka**, **Docker**, and **Kubernetes**
- Led a team of 4 developers building legal tech systems for NGOs; built a USSD service communicating with a Node.js backend via gRPC
- Developed a cross-platform mobile app (React Native, Expo) with Firebase push notifications

PROJECTS

Real-Time Pub/Sub Sync Pipeline

Live MongoDB ↔ Firestore synchronisation pipeline supporting parallel production traffic during a zero-downtime replatform — idempotent, event-driven message processing.

Node.js · Google Pub/Sub · MongoDB · Firestore

Marketplace Recommendation Engine

Collaborative and content-based filtering engine in MongoDB for service-provider matching.

Node.js · MongoDB

Healthcare WhatsApp Chatbot

Integrated OpenAI and Google Vision for prescription OCR via the WhatsApp Business API.

Node.js · OpenAI API · Google Vision

EDUCATION

BSc Computer Engineering — [Makerere University](#), Kampala, Uganda

2018 – 2022

SKILLS

Proficient: Node.js, Python, Firebase, GCP, MongoDB, SQL, React, Git, Linux, HTML, CSS **Intermediate:** Go, AWS, Rust, React Native

Practices: Event-Driven Architecture, Microservices, Docker, Kubernetes, gRPC **Languages:** English, Kiswahili